Aneesh Maganti

aneesh.maganti@nyu.edu • github.com/aminoa • linkedin.com/in/aneesh-maganti

EXPERIENCE

Affirm, New York City, NY, *Software Engineer, Traffic Engineering* July 2025 - Present

• Built a self-serve secrets decryption service via on AWS Lambda driven by an agentic workflow to parse and extract secret metadata, reimplementing three-layer envelope decryption (KEK/DEK) with separate IAM ReaderRoles; automated 40 requests/month, eliminating 60 hours/year of toil.

• Rewrote a core Kubernetes manifest build tool in Rust using agentic workflow, slashing Helm chart build times by 78% (from 14 minutes to 3 minutes); successfully drove team-wide adoption of the tool.

• Orchestrated the migration of External Secrets Operator (ESO) from v0.7.2 to v0.20.2 across 72 Kubernetes clusters, leveraging a Controller Class mechanism to run concurrent operators and manage diverse secret types.

Qualcomm, San Diego, CA, *Software Engineer Intern, AI Software/ML Group* May 2024 - Aug 2024

• Implemented various memory optimizations across the Qualcomm AI Runtime (QAIRT), reducing peak memory usage by 20% across benchmarked ResNet50 and LLAMA2 and LLAMA3 LLM models

• Optimized AIMET quantization pipeline via C++ PyBind API to reference static graph tensors within Python subprocess, reducing in-memory allocations by 50%

• Refactored QAIRT/AIMET toolchain via PyTorch 'meta' device to prevent unnecessary weight initialization and modified model export to use in-memory IR graph, reducing disk space and improving runtime

New York University, Brooklyn, NY, *Teaching Assistant (Machine Learning)* Sep 2023 - Dec 2023

• Instructed students weekly for machine learning topics of written and programming tasks through office hours

• Graded weekly assignments on the basis of proper algorithm implementation, code correctness and style

NYU Algorithms and Foundations Group, Brooklyn, NY, *ML Academic Researcher* Feb 2023 - Sep 2023

• Designed and implemented *Deltagonalshift*, a diagonal estimator for a dynamic matrix under Professor Christopher Musco to improve neural network optimization based on DeltaShift trace estimation algorithm

• Demonstrated Deltagonalshift was more effective than repeatedly running Hutchinson's diagonal estimator

Corelink, Brooklyn, NY, *Software Engineer Intern* Sep 2021 - May 2022

• Implemented a C++ UDP network packet splitter to enable researchers to bypass Corelink's MTU limit from 20,000 to 64,000 bytes, increasing maximum throughput by 220%

• Designed Next.js/React interview scheduling platform using Auth0 for authentication and MongoDB backend

EDUCATION

New York University, Tandon School of Engineering, Brooklyn, NY Fall 2024

Bachelor of Science, Computer Science GPA: **3.80**

Relevant Courses: Algorithmic Machine Learning, Offensive Security, ML Visualization, Computer Architecture

PROJECTS/ACTIVITIES

Dot Matrix - Game Boy Emulator June 2025 - Present

• Re-architected my C++ Game Boy emulator into idiomatic Rust (CPU, MMU, PPU, APU, cartridge, joypad, and renderer modules), driving the system from a cycle-stepped fetch-execute loop with full interrupt handling and accurate timer/clock emulation, and verified CPU correctness against Blargg's cpu_instrs test ROM suite.

• Built a scanline-based PPU implementing the four-mode LCD state machine (OAM scan, pixel transfer, H-Blank, V-Blank) with background, window, and sprite rendering output through a minifb framebuffer.

• Added MBC1 ROM/RAM bank switching, input, audio, and save states, enabling playback of commercial titles.

BUGS Open Source Club President Sep 2022 - Dec 2023

• Started and led 50+ member club by coordinating biweekly workshop and project coding events to discuss software engineering skills, foster contributions to open source, and create a fun, inclusive CS community.

• Led multiple workshops including discussion of open-source licenses, JavaScript Playwright automation, and an overview of emulation and the internals of C++ Game Boy emulator.

• Developed Next.js-React website NYU Syllabi with Netlify hosting and Docusaurus-based NYU CS Wiki websites